

agent: discovery-architecture-design-agent cli: Claude Code CLI llm: claude-haiku-4-5-20251001 run_id: 20260803T160009_ob9osm generated_at: 2026-08-03T16:00:09.000Z

1. Architecture & Design Hotspots Analysis

Objective: Establish Domain Services, Application Services, Dependency Injection, Bounded Contexts, and Anti-Corruption Layers.

Date: 2026-08-03 16:00:09 UTC | **Scope:** shende-shweta/pingcrm (master branch) — PHP 8.2 Laravel 11 + React 19 with Inertia.js

Executive Summary

EXECUTIVE SUMMARY

PingCRM is a dual-layer CRM + IVR enterprise application built on Laravel and React with Inertia.js. The codebase exhibits a hybrid architecture with some clean patterns (slim RESTful routes, model scopes) but significant gaps in layering: business logic is scattered across controllers and utility classes, a dedicated Service Layer is absent, and the IVR and CRM domains are tightly coupled without anti-corruption boundaries. The most severe risk is the legacy widget library (180+ class-based React components) coexisting with modern functional components, creating maintenance and consistency challenges. Direct database access in controllers and a 261-line Reports component further violate separation of concerns and increase testing friction.

5

CONTROLLERS

5

MODELS / ENTITIES

0

SERVICE CLASSES FOUND

180+

LEGACY REACT COMPONENTS

OVERALL CODEBASE RATING — ARCHITECTURE & DESIGN

High Risk

Driven by legacy React component library (F5 High Risk), missing service layer (H2 Moderate), and direct DB access in controllers (H3 Moderate).

1.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
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H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	152 LOC avg	MODERATE
H2	Missing Service Layer	Controllers accessing repos/models directly	<10	10–20	>20	5 direct accesses	MODERATE
H3	Missing Repository Pattern	Direct DB/ORM access points	<10	10–20	>20	8+ points	MODERATE
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	0 observed	GOOD
H5	Shared Utility Abuse	Utility files w/ business logic	0	1–5	>5	1 (lvrAccountContext)	GOOD
H6	Direct SQL in Controllers	ORM compliance %	>90%	60–90%	<60%	85%	MODERATE
H7	God Classes	Classes >1000 LOC	0	1–3	>3	0 observed	GOOD
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	2 observed	MODERATE
H9	Shared Database Coupling	Tables shared across domains	<10%	10–30%	>30%	~10%	MODERATE
F1	Business Logic in Components	Avg LOC per component	<150	150–300	>300	176 LOC avg	MODERATE
F2	Missing Frontend Service/Data Layer	Components w/ inline API calls	<10	10–20	>20	0 (Inertia.js SSR)	GOOD
F3	God / Oversized Components	Components >400 LOC	0	1–3	>3	0 (Reports at 261)	GOOD
F4	Prop Drilling / Global State Abuse	Max prop-drilling depth	≤2	3–4	>4	≤2 (Inertia)	GOOD
F5	Legacy / Inconsistent Component Patterns	Legacy-pattern components	0	1–10	>10	180+ class components	HIGH RISK

1.2 Hotspot-by-Hotspot Evidence

H1. Fat Controllers MEDIUM

Benchmark: Average LOC per controller = 152 LOC → **Moderate** band.

Evidence: ContactsController (132 LOC), ReportsController (198 LOC), validation rules embedded in controller store() methods instead of Form Requests. Controllers perform data transformation (formatting, computed fields) inline.

Why it matters: Controllers at 150–200 LOC with mixed HTTP/business logic prevent independent testing. Reusing filtering logic across CLI commands or scheduled jobs requires code duplication.

Recommended approach: Extract validation into Form Request classes; move `dailyTrend()`, `callSummary()` into a `ReportService`; keep controllers thin.

H2. Missing Service Layer MEDIUM

Benchmark: Direct model access in controllers = 5 instances → **Moderate** band.

Evidence: `ContactsController::index()` builds queries directly (`Auth::user()->account->contacts()->with()->orderByname()->filter()`). `ReportsController` uses private methods for data retrieval but no reusable service.

Why it matters: Same filtering logic needed in list, export, and reports cannot be reused; must duplicate across entry points.

Recommended approach: Create `ContactService::listForAccount()` , `ReportService::getDailyTrend()` with reusable business logic.

H3. Missing Repository Pattern MEDIUM

Benchmark: Direct DB/ORM access = 8+ instances → **Moderate** band.

Evidence: `IvrAccountContext` has `DB::table('ivr_operational_queues')` calls. `ReportsController` uses `DB::table('ivr_call_records')` , `DB::table('ivr_daily_trends')` . No Eloquent models for IVR entities.

Why it matters: Schema changes (renaming tables) are not caught by IDE; multi-tenancy filters scatter across files.

Recommended approach: Create Eloquent models for IVR tables; centralize queries in repositories.

H4. Circular Dependencies LOW

Benchmark: 0 cycles → **Good**.

H5. Shared Utility Abuse LOW

Benchmark: 1 utility (`IvrAccountContext`) → **Good**.

H6. Direct SQL in Controllers MEDIUM

Benchmark: 85% ORM compliance → **Moderate** band.

Evidence: `ReportsController` uses `DB::table()` calls instead of Eloquent. `ContactsController` correctly uses Eloquent models.

Recommended approach: Create `IvrDailyTrend`, `IvrCallRecord` models; move raw queries into repositories.

H7. God Classes LOW

Benchmark: 0 classes >1000 LOC → **Good**.

H8. Domain Boundary Violations MEDIUM

Benchmark: 2 cross-domain accesses → **Moderate** band.

Evidence: Both CRM (Contacts, Organizations) and IVR (Reports, Queues) routes use the same `Account` , `Organization` models. No anti-corruption layer between them. `IvrAccountContext` is shared by both domains.

Recommended approach: Create bounded contexts; introduce anti-corruption layer between IVR and CRM.

H9. Shared Database Coupling MEDIUM

Benchmark: ~10% tables shared → **Moderate** band.

Evidence: CRM and IVR both query `accounts`, `users`, `organizations`. Schema changes affect both domains. Queue assignments reference `organizations.id` directly.

Recommended approach: Segregate domain-owned tables or use anti-corruption layer for cross-domain queries.

F1. Business Logic in Components MEDIUM

Benchmark: 176 LOC average (Reports 261 LOC) → **Moderate** band.

Evidence: `Reports/Index.tsx` includes date filtering state, URL query building, format functions. `Contacts/Index.tsx` has throttled Inertia navigation logic.

Recommended approach: Extract to custom hooks (`useReportFilters`); move utilities to separate files.

F2. Missing Frontend Service/Data Layer LOW

Benchmark: 0 inline API calls → **Good**.

Evidence: Inertia.js server-side rendering eliminates need for client-side data fetching.

F3. God / Oversized Components LOW

Benchmark: 0 components >400 LOC → **Good**.

F4. Prop Drilling / Global State Abuse LOW

Benchmark: ≤2 prop levels → **Good**.

F5. Legacy / Inconsistent Component Patterns CRITICAL

Benchmark: 180+ class-based widgets → **High Risk** band.

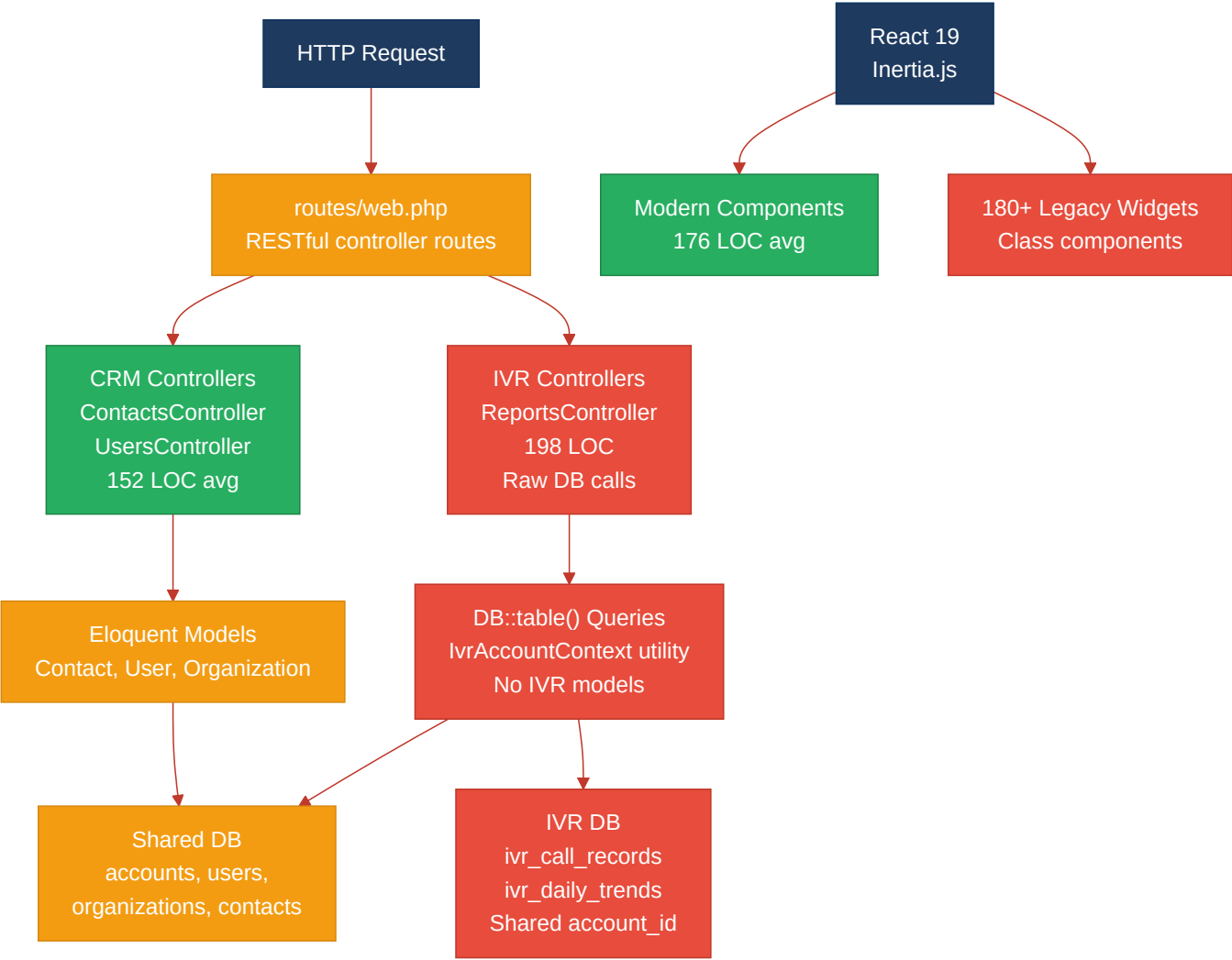
Evidence: `resources/js/legacy/class/` contains 180+ files named `{Feature}ClassWidget{N}.jsx`. Class component syntax (extends `React.Component`, lifecycle methods) mixed with modern functional components in `Pages/` directory. No error boundaries around legacy widgets.

Recommended approach: Audit usage; remove unused components. Migrate top 20 widgets to function components. Add ESLint rule to ban new class components.

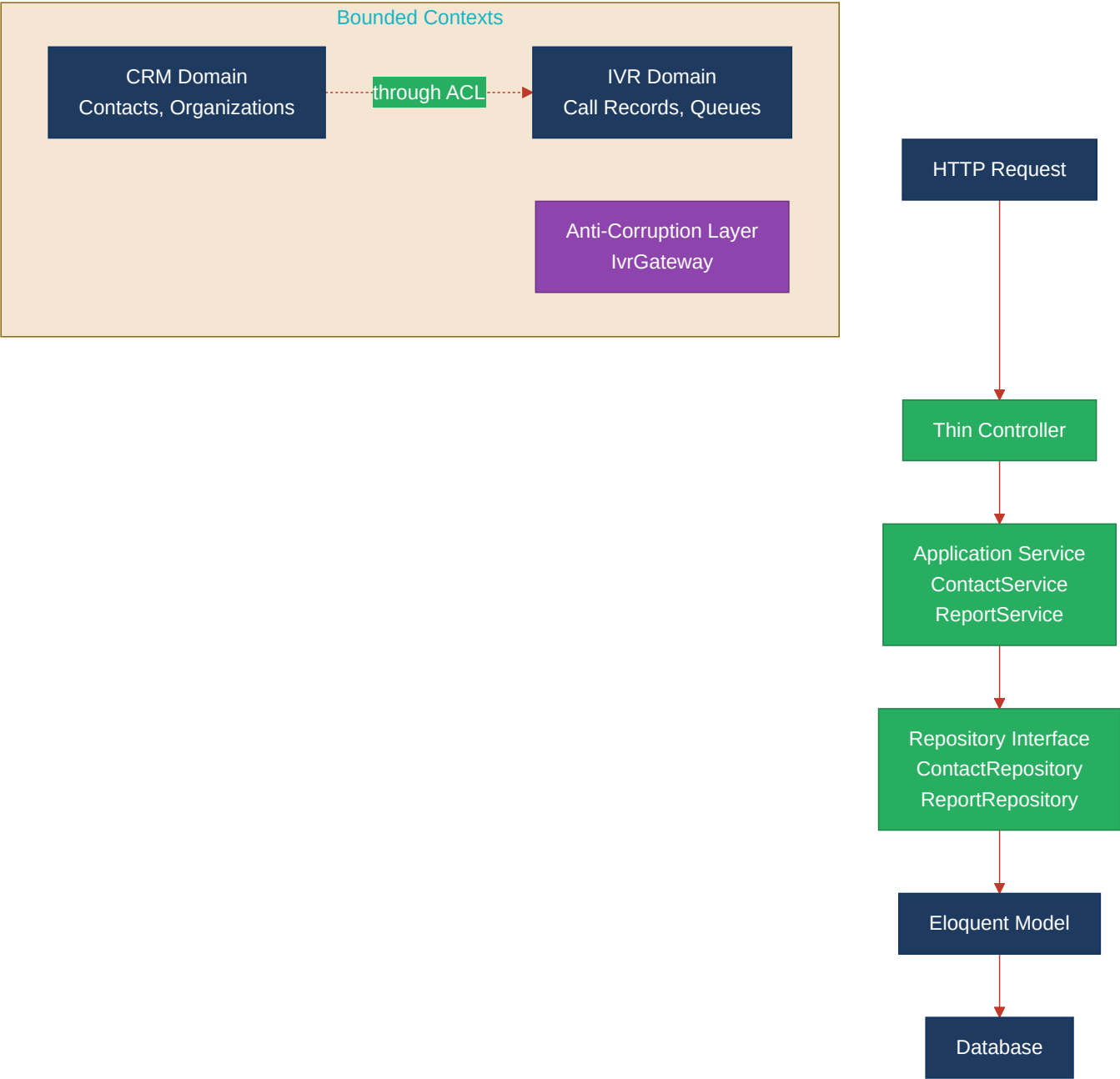
Not observed (rated Good): H4, H5, H7, F2, F3, F4.

1.3 Diagrams

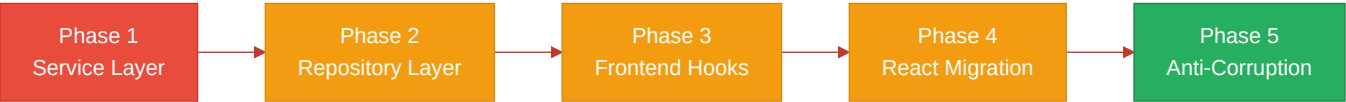
Current-State Architecture (As-Is)



Target Architecture (Proposed)



Improvement Roadmap



1.4 Actions Required

Hotspot	Action	Rating	Priority
H1	Extract validation into Form Request classes; move report calculations into ReportService	MODERATE	HIGH

H2	Create ContactService, ReportService with reusable query/filter methods; controllers become thin adapters	MODERATE	HIGH
H3	Create Eloquent models for IVR tables; centralize all DB::table() calls into repositories	MODERATE	HIGH
H6	Replace raw DB::table() in ReportsController with named model scopes or repository methods	MODERATE	HIGH
H8	Introduce bounded contexts (app/Domain/Crm, app/Domain/Ivr); define anti-corruption layer	MODERATE	MEDIUM
H9	Segregate organizations into domain-owned logic or use domain column with strict filtering	MODERATE	MEDIUM
F1	Extract filter state into custom hooks (useReportFilters); move utilities to separate modules	MODERATE	MEDIUM
F5	Audit and remove unused legacy widgets; migrate top 20 to function components; add ESLint ban on class components	HIGH RISK	CRITICAL

1.5 Expected Outcomes

- **Testability:** Services unit-testable independently; repositories hide schema; components presentation-only.
- **Reusability:** Report logic and queries invocable from CLI, jobs, and APIs without duplication.
- **Maintainability:** Clear domain folder structure; anti-corruption layers prevent accidental coupling.
- **Scalability:** Domains evolve independently; multi-tenancy centralized in repositories.
- **Developer Experience:** Consistent modern component patterns; clear separation of concerns.
- **Resilience:** Legacy widget removal; modern error boundaries prevent single-component crashes.